

Project Executable Files

Date	15 March 2026
Team ID	
Project Name	Smart Lender:AI/ML-Based Loan Approval Prediction System

Project Executable Files

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA (<i>No database used</i>)
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes / NA (based on my deployment)
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA (<i>Web application</i>)
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

```
Smart_Lender/
|
├── app.py          # Flask web application (main backend)
├── model.pkl       # Trained XGBoost ML model
├── preprocess.py   # Data preprocessing
script ├── train_model.py # Model training
script |
├── requirements.txt # Python dependencies
├── README.md       # Setup and run instructions
├── .env           # Environment variables (if used)
|
├── templates/     # HTML files (frontend)
|   ├── index.html # Input form page
|   └── result.html # Prediction result page
|
├── static/        # CSS, JS, images
|   ├── css/
|   |   └── style.css
|   ├── js/
|   |   └── script.js
|
├── dataset/       # Dataset used for training
|   └── loan_data.csv
|
├── tests/         # Performance testing evidence
|   ├── jmeter_report.png
|   ├── locust_graph.png
|   └── k6_output.png
|
├── screenshots/   # UI screenshots of project
|   ├── home_page.png
|   └── prediction_page.png
|
└── docs/          # Documentation files
    ├── project_report.pdf
    └── architecture_diagram.png
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	http://127.0.0.1:5000/
Login Credentials (Demo)	https://github.com/sowmya-123456/sowmya-123AI-ML-Track-smartlender-Project-Template.git
Platform / Hosting Provider	Local Machine (Flask Development Server)
Repository Link	https://github.com/sowmya-123456/sowmya-123AI-ML-Track-smartlender-Project-Template.git
Demo Video Link	https://drive.google.com/file/d/1CBDs4Upj_MNpmEOmPLENGiT5ACeVqu4W/view?usp=drivesdk

Step 4: Run Instructions

Pre-requisites

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB minimum (8 GB recommended)
- Storage: 10 GB free space
- 64-bit Operating System

Software Requirements

- Windows 10/11, Linux, or macOS
- Python 3.8 or above
- Anaconda Navigator (optional)
- Visual Studio Code or PyCharm
- Google Chrome or Microsoft Edge
- Git (optional, for cloning the repository)

Step 1: Clone the Repository

```
git clone https://github.com/sowmya-123456/sowmya-123-AI-ML-Track-smartlender-ProjectTemplate.git
```

Step 2: Navigate to the Project Folder

```
cd Smart-
```

Step 3: Install Required Dependencies

```
pip install -r requirements.txt
```

If `requirements.txt` is unavailable, install the required packages manually:

```
pip install flask pandas numpy scikit-learn matplotlib seaborn xgboost joblib
```

Step 4: Verify the Model File

Ensure the trained model file (`model.pkl`) is available in the project directory.

Step 5: Run the Flask Application

```
python app.py
```

Step 6: Open the Application

<http://127.0.0.1:5000/> or
<http://localhost:5000/>

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Prediction accuracy depends on the quality and completeness of the input data.	Ensure all required fields are filled with valid values before prediction.
2	The model is trained on historical loan data and may not generalize well to unseen or significantly different datasets.	Retrain the model periodically using updated loan application data.
3	The application supports only single loan prediction through the web interface and does not process bulk applications.	Future enhancement: Add batch prediction using CSV file upload.